

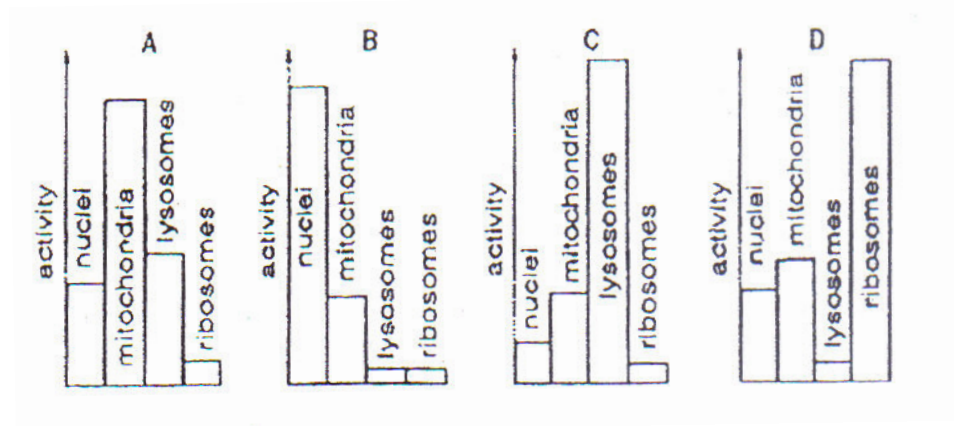
- 1 A microscope is set up to examine the surface of a leaf. The 40x objective is in position and a 7x eyepiece is being used. The diameter of the field of view is 0.4mm.

What is the approximate density of the stomata if 15 of them are visible under these conditions?

- A 33 stomata/mm²
- B 120 stomata/mm²
- C 528 stomata/mm²
- D 1680 stomata/mm²

- 2 A piece of mammalian tissue was homogenized and subjected to differential centrifugation. Four subcellular fractions were investigated to determine their relative biochemical activity. The diagrams indicate the relative activity of certain biochemical processes in these fractions.

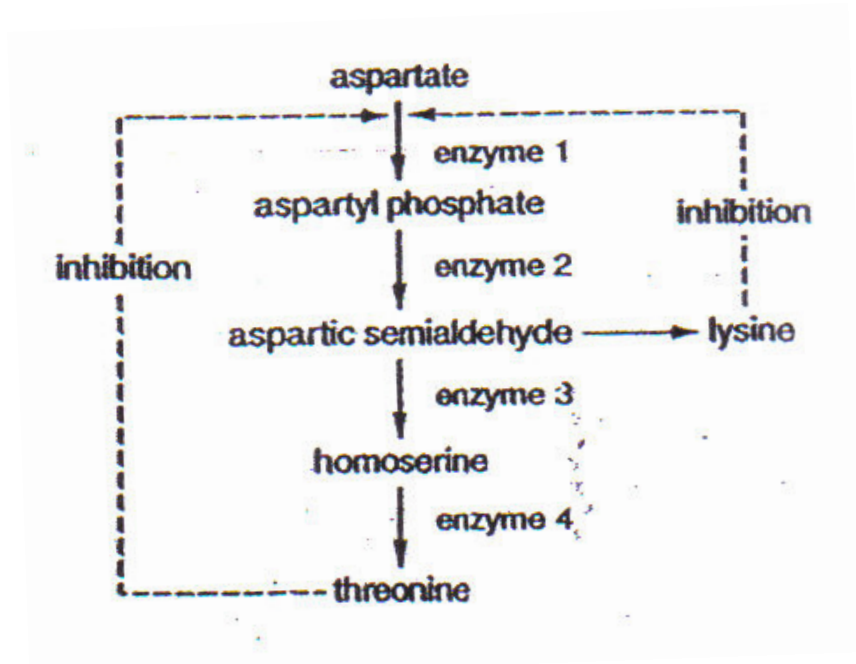
Which diagram indicates the fraction with maximum synthesis of messenger RNA?



- 3 What is the main function of the Golgi apparatus?
- A Formation of glycoprotein
 - B Phosphorylation of ADP
 - C Synthesis of proteins
 - D Transport of RNA to the nucleus
- 4 Which of the following is a characteristic of saturated fats which distinguishes them from unsaturated fats?
- A They have no double bonds between their carbon atoms.
 - B They are composed only of fatty acids and glycerol.
 - C They contain a high proportion by mass of oxygen.
 - D They contain only unbranched fatty acids.

- 5 A culture of bacteria produces the food supplement lysine by the metabolic pathway shown below.

Which change in enzyme activity will result in the greatest increase in lysine yield?



	Enzyme	Change in activity
A	1	Decrease
B	2	Increase
C	3	Increase
D	4	Decrease

- 6 The following events occur in mitosis and/or meiosis.

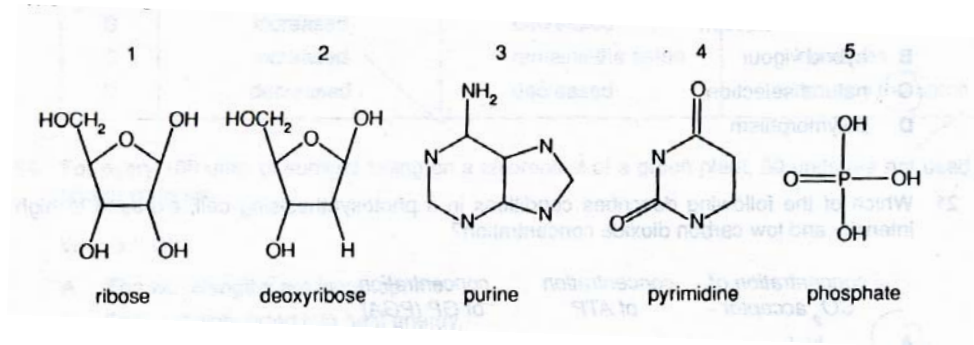
Which event occurs only in meiosis?

- A Chromatid formation
- B Chromosome condensation
- C Chromosome movement to poles
- D Chromosome pairing

- 7 What is a definition of homologous chromosomes?

- A They are divided into two identical chromatids
- B They have the same sequence of bases.
- C They have the same sequence of genes.
- D They move to the same pole during nuclear division.

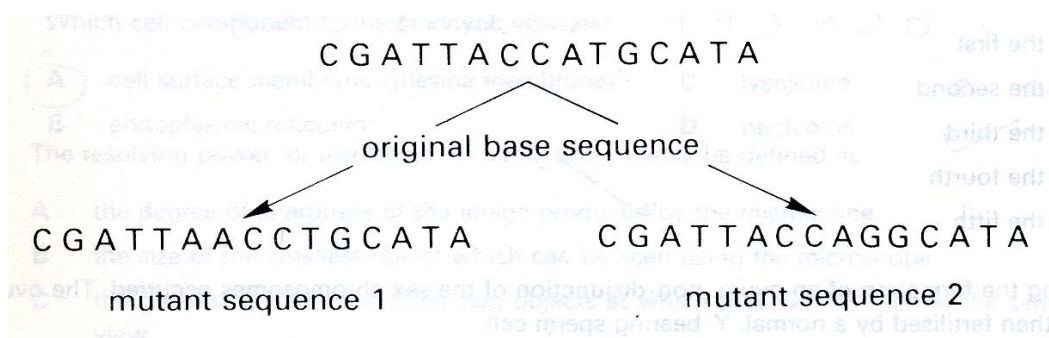
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Which of these molecules are linked to form a nucleotide containing uracil?

- A 1,3 and 5
- B 1,4 and 5
- C 2,3 and 5
- D 2,4 and 5

9 The diagram shows two types of gene mutation



What types of gene mutation produce a mutant sequence 1 and 2?

- | | 1 | 2 |
|---|--------------|--------------|
| A | insertion | deletion |
| B | inversion | substitution |
| C | substitution | insertion |
| D | substitution | inversion |

10 If the nucleus of a hamster sperm cell contains 3.4 picograms (pg) of DNA, what mass of DNA is the nucleus of an actively dividing hamster skin cell likely to contain at the end of interphase?

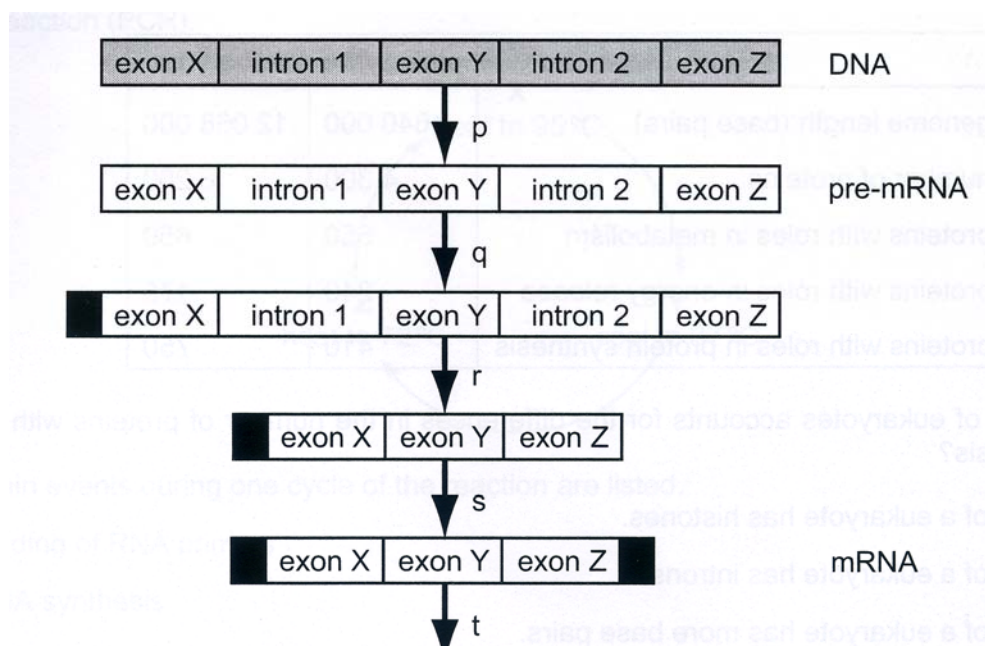
- A 3.4 pg
- B 6.8pg
- C 13.6 pg
- D 27.2 pg

11 Which of the listed features are characteristic of a prokaryotic genome?

- 1 polycistronic mRNA
- 2 DNA associated with histones
- 3 transcription by 80S ribosomes
- 4 group of structural genes clustered together

- A** 1 only
B 3 only
C 1 and 4
D 3 and 4

12 The diagram represents the processing of pre-messenger RNA.

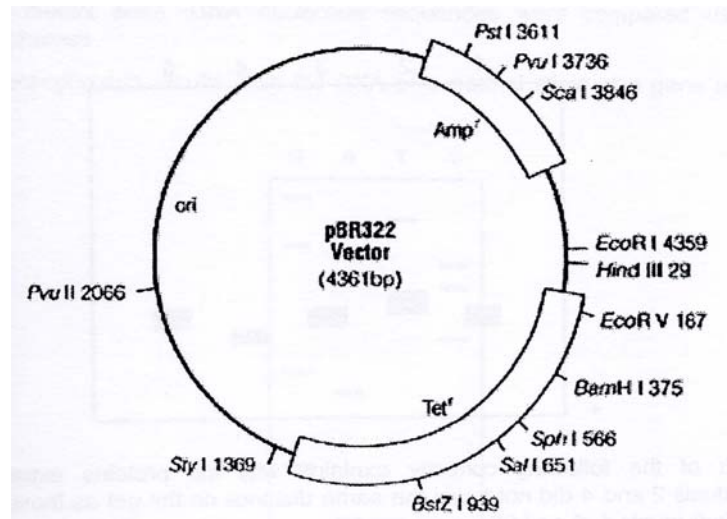


During the process of pre-messenger RNA each of the listed events occurs.

- 1 capping
- 2 polyadenylation
- 3 splicing
- 4 transcription
- 5 translation

- | | p | q | r | s | t |
|----------|----------|----------|----------|----------|----------|
| A | 5 | 1 | 3 | 2 | 4 |
| B | 4 | 1 | 3 | 2 | 5 |
| C | 5 | 2 | 1 | 3 | 4 |
| D | 4 | 2 | 1 | 3 | 5 |

- 13 When glucose is absent and lactose is present, lactose binds to the operon's repressor. Which statement best describes what happens subsequently?
- A** Repressor binds to promoter, transcription is initiated.
 - B** Repressor binds to operator, transcription is not initiated.
 - C** Repressor fails to bind to structural genes, transcription is not initiated.
 - D** Repressor fails to bind to the operator, transcription is initiated.
- 14 pBR322 vector is used to clone a eukaryotic gene which has been digested by the restriction endonuclease *Bam*HI.



Following transformation, bacterial cells were grown in four different media, as shown below:

- 1 nutrient medium + ampicillin
- 2 nutrient medium + tetracycline
- 3 nutrient medium + without antibiotics
- 4 nutrient medium + ampicillin + tetracycline

Which of the following media would bacterial cells contain the recombinant plasmids grow in?

- A** 1 and 2
 - B** 1 and 3
 - C** 2 and 3
 - D** 2 and 4
- 15 In *Drosophila*, ebony body colour and vestigial wings are recessive to the wild type and are not linked. Crossing two flies heterozygous for both characters produced 256 offspring.

How many offsprings would be expected to have ebony body colour with normal wings?

- A** 6
- B** 48
- C** 96
- D** 144

- 16 Two animals are mated. One is homozygous dominant for one character and homozygous recessive for another. The other animal is heterozygous for both characters.

How many phenotypes are expected in the offspring of this cross?

- A** 1
- B** 2
- C** 3
- D** 4

- 17 A pure breeding plant with red flowers and tall stems was crossed with one with white flowers and short stems. The F_1 plants all had pink flowers and tall stems. These were interbred to produce an F_2 generation. The following ratios of plant phenotypes were obtained in this generation.

3/16	red flowers tall plants
1/16	red flowers short plants
3/8	pink flowers tall plants
1/8	pink flowers short plants
3/16	white flowers tall plants
1/16	white flowers short plants

Which of the following statements is true with respect to inheritance of flower colour and plant height?

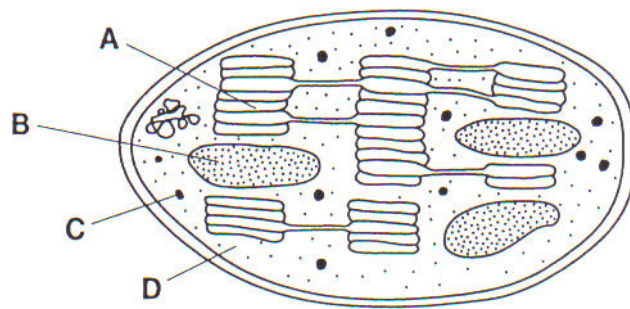
- A** Red flowers and tall stature are both dominant characteristics.
 - B** White flowers and short stature are both dominant characteristics.
 - C** Red flower colour is incompletely dominant to white flower colour and tall stature is dominant to short stature.
 - D** Red flower colour is incompletely dominant to white colour and tall stature is incompletely dominant to short stature.
- 18 Which of the following increases the number of different alleles in a population?
- A** Crossing over
 - B** Gene mutation
 - C** Random fusion of gametes
 - D** Reassortment of chromosomes in meiosis

- 19 A strain of toad has only one nucleolus in the nucleus of each cell instead of the usual two. When toads with one nucleolus per cell are mated, approximately a quarter of the offspring have two nucleoli per nucleus, half have one nucleolus per nucleus and a quarter have no nucleoli.

What is the most likely explanation of these results?

- A** The possession of one nucleolus is due to autosomal linkage.
- B** The possession of one nucleolus is due to the heterozygous condition.
- C** The allele for the presence of two nucleoli is recessive.
- D** The allele for the presence of two nucleoli is dominant.

- 20 The diagram represents a section through a chloroplast.



In which part does chemiosmosis takes place?

- 21 From yeast cells, a mixture of enzymes can be obtained that catalyses the conversion of glucose to carbon dioxide and ethanol. Pyruvate is produced during this process.

Which combination of substances would increase the rate of ethanol production by these enzymes?

- A** ATP and lactate
- B** ATP and NAD
- C** NAD and coenzyme A
- D** NAD and lactate

- 22 The concentration of carbon dioxide in a sample of air was found to be 280 ppm (parts per million). A controlled experiment was designed to measure the concentration of carbon dioxide in the air after it had flowed over the leaves of a green plant. Measurements were taken at a range of light intensities.

The following results were obtained:

Light intensity (% of full sunlight)	Concentration of carbon dioxide in air after flowing over leaves (ppm)
75	253
50	252
25	254
10	280

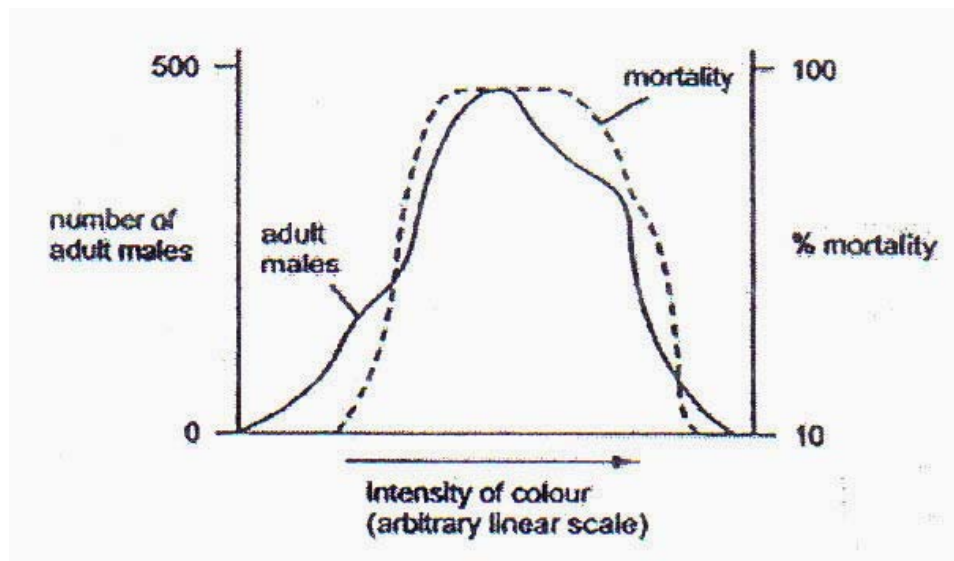
Which one of the following statements is not consistent with these results?

- A** At the lower light intensities tested, the rate of photosynthesis is limited by light intensity.
 - B** At the higher light intensities tested, the rate of photosynthesis is affected by factors other than light.
 - C** In the dark, the concentration of carbon dioxide in the air after it had flowed over leaves would be at least 280 ppm.
 - D** At a light intensity of 10% of full sunlight, photosynthesis does not occur.
- 23 Some organisms change the composition of their membranes to maintain membrane fluidity when temperature changes.

Which changes occur in the composition of membranes following a change to low temperature?

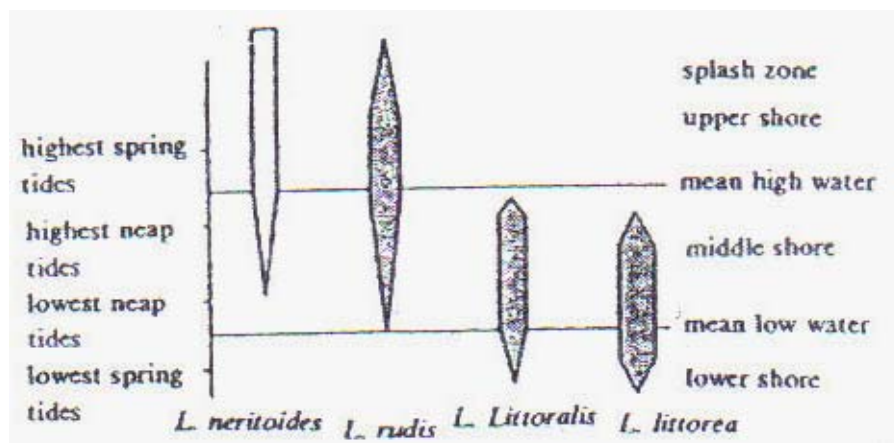
- A** Increase phospholipids, decrease cholesterol
 - B** Increase saturated fatty acids, increase cholesterol
 - C** Increase saturated fatty acids, decrease cholesterol
 - D** Increase unsaturated fatty acids, increase cholesterol
- 24 Within a few weeks of treatment with an anti-viral drug, a patient's viral population in the blood stream consists entirely of drug-resistant viruses. How can this be best explained.
- A** The patient must have been re-infected with drug-resistant viruses.
 - B** The viruses began making drug-resistant offspring in response to the drug.
 - C** Some viruses developed drug resistance and then passed their resistance gene to all viruses present in the patient.
 - D** A few drug resistant viruses were present at the start of the treatment and natural selection increase their frequency.

- 25 The graph below shows data on a population of a species of moth which shows considerable variation in colour intensity



Which conclusion can be made from this graph?

- A Colour variation is environmentally induced.
 - B Colour variation is genetically determined.
 - C Extreme forms are favoured by natural selection.
 - D The species shows discontinuous variation with respect to colour.
- 26 The diagram below shows the frequency and distribution of four Littoral species (periwinkles) on a rocky shore. All feed in a snail-like manner by grazing on algae.

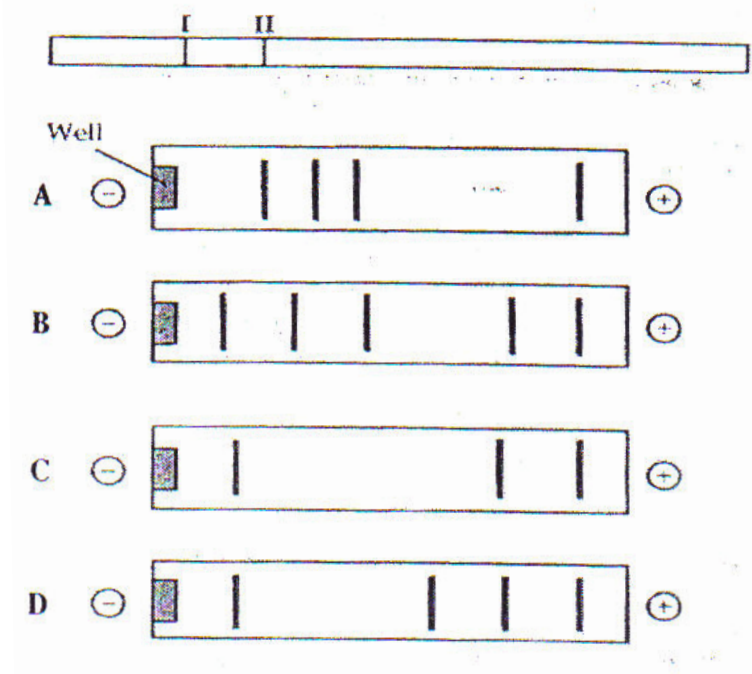


Which of the following factors could not directly contribute to this distribution pattern?

- A Variation in the tolerance of each species to desiccation.
- B Variation in the tolerance of each species to wave action.
- C Photoperiod and seasonal change in day length.
- D Differential selection of periwinkles by predators.

- 27 The restriction fragment shown below contains a gene whose recessive allele is lethal. The normal allele has restriction sites for restriction enzyme *Pst*I at sites I and II. The recessive allele lacks restriction site I. An individual who had a sister with the lethal trait is being tested to determine if he is a carrier of that lethal.

Which of the band patterns would be produced on a gel if he is a carrier?



- 28 Expression of a cloned eukaryotic gene in a prokaryotic cell involves many difficulties. The use of mRNA and reverse transcriptase is part of a strategy to solve the problem of _____.
- A** post-transcriptional processing
 - B** post-translational processing
 - C** nucleic acid hybridization
 - D** restriction fragment ligation
- 29 Which of the following statements regarding stem cells is false?
- A** One potential side effect of any embryonic stem cell-based therapy is the formation of tumour.
 - B** Stem cells are present within various organs of the adult body.
 - C** Stem cells can develop into a whole organism when implanted into the womb.
 - D** Stem cells can be grown and expanded indefinitely in culture under appropriate culture conditions.

- 30 Which statement best describes the characteristics of a pluripotent cell?

- A** Cells that can differentiate into embryonic and extra embryonic cell types.
- B** Cells that can differentiate into all the cells found in various tissues of the body.
- C** Cells that can only produce closely related family of cells.
- D** Cells that can produce only one cell type but have the property of self-renewal.

End of Paper